

Prime & Composite Numbers

Number World Study Notes

Definitions, primality testing, and prime factorization

Prime numbers are the 'atoms' of arithmetic — every whole number greater than 1 can be broken down into a unique product of primes. Understanding primes and composites helps with factorization, simplifying fractions, and finding LCM/HCF.

Key Definitions

Prime Number

A whole number greater than 1 that has exactly two factors: 1 and itself. Examples: 2, 3, 5, 7, 11, 13, ...

Composite Number

A whole number greater than 1 that has more than two factors. Examples: 4 (1, 2, 4), 6 (1, 2, 3, 6), 9 (1, 3, 9).

Special Case: 1

The number 1 has only one factor (itself), so it is classified as neither prime nor composite.

Prime Factorization

Writing a composite number as a product of only prime numbers, e.g., $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$.

Key Facts & the Primality Test

- 2 is the smallest prime number and the only even prime — every other even number is divisible by 2, so it is composite.
- Primes under 100 (25 of them): 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.
- To test if n is prime, try dividing it by every prime number up to $\sqrt{n}$. If none divide evenly, n is prime.
- A factor tree repeatedly splits a number into two factors until every branch ends in a prime.
- The Fundamental Theorem of Arithmetic: every composite number has exactly one prime factorization (ignoring order).

Worked Examples

Example 1: Is 91 a prime number?

$\sqrt{91}$ is approximately 9.5, so test primes up to 9: 2, 3, 5, 7.

91 is odd (not divisible by 2); digit sum 10 is not divisible by 3; does not end in 0/5.

Check 7: $91 \div 7 = 13$ exactly.

Answer: Not prime (composite); $91 = 7 \times 13$

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Example 2: Draw a factor tree for 84 and find its prime factorization.

$$84 = 2 \times 42$$

$$42 = 2 \times 21$$

$$21 = 3 \times 7$$

All branches now end in primes: 2, 2, 3, 7.

Answer: $84 = 2^2 \times 3 \times 7$

Example 3: Is 2 a prime number, and why is it special among primes?

2 has exactly two factors: 1 and 2, so it is prime.

It is the only even prime, because every other even number is divisible by 2 (giving a third factor).

Answer: Yes, 2 is prime; it is the only even prime number

Example 4: Find the prime factorization of 100.

$$100 = 2 \times 50$$

$$50 = 2 \times 25$$

$$25 = 5 \times 5$$

Answer: $100 = 2^2 \times 5^2$

Practice Questions

- 1 List all prime numbers between 40 and 60.
- 2 Is 51 prime or composite? Show your work.
- 3 Find the prime factorization of 72.
- 4 Find the smallest 3-digit prime number.
- 5 How many prime numbers are there between 1 and 20?

Answer Key

- 1 41, 43, 47, 53, 59.
- 2 Composite; $51 = 3 \times 17$.
- 3 $72 = 2^3 \times 3^2$.
- 4 101.
- 5 8 primes: 2, 3, 5, 7, 11, 13, 17, 19.